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Department of Artificial Intelligence (AI) and Data Science

A Presentation on

"A Decentralized IoT-Blockchain Framework with Customized Smart Contracts for Transparent and Traceable Agricultural Supply Chains"

Presented By

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Under the Guidance of
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Agenda

- Introduction
- Problem Statement
- Objectives
- Scope
- Literature Survey
- Methodology & Flow Chart of Project Work
- Software & Hardware Requirement Specification
- Project Scheduling / Timelines
- E-R Diagram
- Use Case Diagram
- Design and Implementation
- Conclusion
- References

Introduction

- **Agricultural supply chains** involve multiple stakeholders, making product origin, quality, transactions, and farmer payments difficult to track transparently.
- **IoT sensors and blockchain** can provide real-time data and immutable records, but blockchain alone cannot verify whether the incoming sensor data is trustworthy.
- Our proposed system introduces an **AI Trust Layer** to validate IoT data before recording it on the blockchain, while **customized smart contracts** manage traceability, custody, quality, and payment-related transactions.
- The system aims to achieve **transparent, trustworthy, end-to-end agricultural traceability** while improving **payment transparency and fairness for farmers**.

Problem Statement

Traditional agricultural supply chains rely on fragmented, siloed record-keeping that leaves farmers, processors, distributors, retailers and consumers without real-time visibility or trust in product origin and quality. This lack of transparency enables food fraud, delayed recalls and poor accountability, while existing blockchain-based solutions remain proof-of-concept, still facing scalability, interoperability, smart-contract security and off-chain data-trust challenges. There is a need to design and implement a scalable, secure blockchain system with customized smart contracts that delivers real, end-to-end traceability across the agricultural supply chain.

Objectives

1. To study existing research papers, identify their contributions, and analyze the research gaps to define the need for the proposed system.

2. To design and develop a blockchain-based framework for the agricultural supply chain that records every transaction (farm, processing, distribution, retail) as an immutable, traceable ledger entry.

3. To implement customized smart contracts that automate stakeholder-specific rules such as quality checks, payment release, and ownership transfer without relying on manual or intermediary verification.

4. To integrate IoT-based sensing devices with the blockchain layer for automated, real-time capture of data (e.g., temperature, location, handling conditions) and validate this data before it is committed on-chain.

5. To build and test the system's performance in terms of scalability, latency, throughput, security and cost-efficiency under realistic transaction loads.

6. To evaluate the implemented system against existing approaches and refine it to address identified gaps such as smart-contract vulnerabilities, interoperability with ERP/IoT platforms, and trust in oracle-fed data.

Scope

- Farmer-centric agricultural supply chain
- IoT-based crop and environmental data collection
- AI-based sensor data validation
- Blockchain-based produce traceability
- Customized smart contracts
- Product and custody tracking
- Transparent transaction records
- Farmer payment transparency
- QR-based farm-to-consumer verification

Literature Survey

Ref No	Paper Title	Author(s) & Year	Focus	Key Contribution	Limitation
1	**Blockchain Technology for Global Supply Chain Management: A Survey of Applications, Challenges, Opportunities and Implications**	Dudczyk, Dunston & Crosby (2024)	Blockchain in Global Supply Chains	Reviews blockchain applications and benefits for global supply-chain transparency and traceability.	Lacks agriculture-specific evaluation.
2	**Revolutionizing Agri-Food Supply Chain Management with Blockchain-Based Traceability and Navigation Integration**	Aggarwal, Rani, Rani & Sharma (2024)	Blockchain in Agri-food	Provides blockchain-based traceability and navigation for food provenance and consumer trust.	Scalability concerns in large-scale deployment.
3	**High-Efficiency Blockchain-Based Supply Chain Traceability**	Wu, Jiang & Cao (2023)	Blockchain Traceability	Improves efficiency of blockchain-based supply-chain traceability by reducing search latency and storage overhead.	Limited real-world/large-scale validation.
4	**Integrating Blockchain Technology for Enhanced Transparency and Security in Supply Chains**	Chaker & Damak (2024)	Blockchain for Supply Chain	Discusses blockchain integration for supply-chain transparency and security.	Mostly conceptual; limited quantitative evidence.
5	**Demystifying the Impact of Blockchain on Trust in Emerging and Established Relationships: A Case of Organic Food Supply Chains**	Yavaprabhas, Kurnia, Seyedghorban & Samson (2024)	Blockchain in Organic Food	Examines how blockchain can improve trust in organic food supply-chain relationships.	Findings are context-specific to organic food.

Ref No	Paper Title	Author(s) & Year	Focus	Key Contribution	Limitation
1	**Optimizing Supply Chain Management With IoE and AI**	Agrawal, Aronkar, Palav, Badre, Karumuri & Bagale (2025)	IoE & AI in SCM	Explores AI and Internet of Everything for supply-chain optimization and decision-making.	Lacks empirical validation in agricultural contexts.
2	**Zero-Day Exploits Detection with Adaptive WavePCA-Autoencoder (AWPA) Adaptive Hybrid Exploit Detection Network (AHEDNet)**	Mohamed, Al-Saleh, Sharma & Tejani (2025)	ML-Based Security	Uses a hybrid ML approach for proactive zero-day exploit detection.	Requires high-quality, large training datasets.
3	**A Novel Malware Detection Model in the Software Supply Chain Based on LSTM and SVMs**	Zhou, Li, Fu & Jiao (2024)	ML Malware Detection	Uses LSTM and SVM-based models for malware detection.	Higher model complexity increases deployment cost.
4	**Harnessing Advanced Hybrid Deep Learning Model for Real-Time Detection and Prevention of Man-in-the-Middle Cyber Attacks**	Kandasamy & Roseline (2025)	Deep Learning Security	Applies hybrid deep learning for real-time MITM attack detection and prevention.	Computationally intensive for constrained devices.

Ref No.	Paper Title	Author(s) & Year	Focus	Key Contribution	Limitation
1	**Secure Pharmaceutical Supply Chain Using Blockchain in IoT Cloud Systems**	Mangala, Naveen, Reddy, Buyya, Venugopal, Iyengar & Patnaik (2024)	Blockchain + IoT	Integrates blockchain with IoT-cloud systems for secure authentication and anti-counterfeiting.	High system complexity limits adoption.
2	**Enhancing Supply Chain Resilience and Efficiency Through Internet of Things Integration: Challenges and Opportunities**	Shoomal, Jahanbakht, Componation & Ozay (2024)	IoT-Enabled SCM	Shows how IoT improves supply-chain resilience, efficiency and coordination.	Data privacy risks remain unresolved.
3	**Supply Chain Management Growth With the Adoption of Blockchain Technology (BoT) and Internet of Things (IoT)**	Singh & Dadhich (2023)	Blockchain + IoT	Presents a model showing improved supply-chain coordination and visibility through Blockchain + IoT.	Lacks real implementation and empirical testing.
4	**Blockchain-Driven Security for IoT Networks: State-of-the-Art, Challenges and Future Directions**	Maurya, Rishiwal, Yadav, Shiblee, Yadav, Agarwal & Chaudhry (2025)	IoT Security	Reviews blockchain-based approaches for improving IoT security and data protection.	Introduces computational overhead.
5	**Enhancing IoT Data Security: Using the Blockchain to Boost Data Integrity and Privacy**	Eghmazi, Ataei, Landry & Chevrette (2024)	IoT Data Security	Uses blockchain to improve IoT data integrity, privacy and security.	Scalability with high sensor-data volume is not fully addressed.

Ref No.	Paper Title	Author(s) & Year	Focus	Key Contribution	Limitation
1	**Revolutionizing Agri-Food Supply Chain Management with Blockchain-Based Traceability and Navigation Integration**	Aggarwal, Rani, Rani & Sharma (2024)	Agri-food Traceability	Improves end-to-end food provenance, traceability and consumer trust.	Scalability concerns in large-scale deployment.
2	**Demystifying the Impact of Blockchain on Trust in Emerging and Established Relationships: A Case of Organic Food Supply Chains**	Yavaprabhas, Kurnia, Seyedghorban & Samson (2024)	Organic Food Supply Chain	Demonstrates blockchain's role in strengthening trust within organic food supply chains.	Findings are specific to organic food case studies.
3	**Optimizing Supply Chain Management With IoE and AI**	Agrawal, Aronkar, Palav, Badre, Karumuri & Bagale (2025)	AI in Supply Chain	Explores AI and IoE for supply-chain optimization.	Lacks empirical validation in agricultural contexts.
4	**Enhancing Supply Chain Resilience and Efficiency Through Internet of Things Integration: Challenges and Opportunities**	Shoomal, Jahanbakht, Componation & Ozay (2024)	IoT Supply Chain	Highlights IoT applications for real-time tracking and improved supply-chain resilience, including agri-food chains.	Privacy, interoperability and scalability challenges.
5	**Blockchain Technology for Global Supply Chain Management: A Survey of Applications, Challenges, Opportunities and Implications**	Dudczyk, Dunston & Crosby (2024)	Global Supply Chain	Identifies blockchain's benefits for transparency and traceability across supply chains.	Does not specifically evaluate agricultural applications.

Proposed Methodology

1. Data Collection: Collect agricultural supply-chain data using IoT sensors and stakeholder inputs.

2. Data Validation: Use the **AI Trust Layer** to validate sensor data and identify abnormal or unreliable readings.

3. Blockchain Recording: Store validated transactions and supply-chain events on a **blockchain** to ensure transparency and tamper resistance.

4. Smart Contracts: Use **customized smart contracts** to automate quality verification, ownership/custody transfer, and payment-related transactions.

5. Traceability: Track the agricultural product from **farmer → intermediaries → market/consumer** with a verifiable digital record.

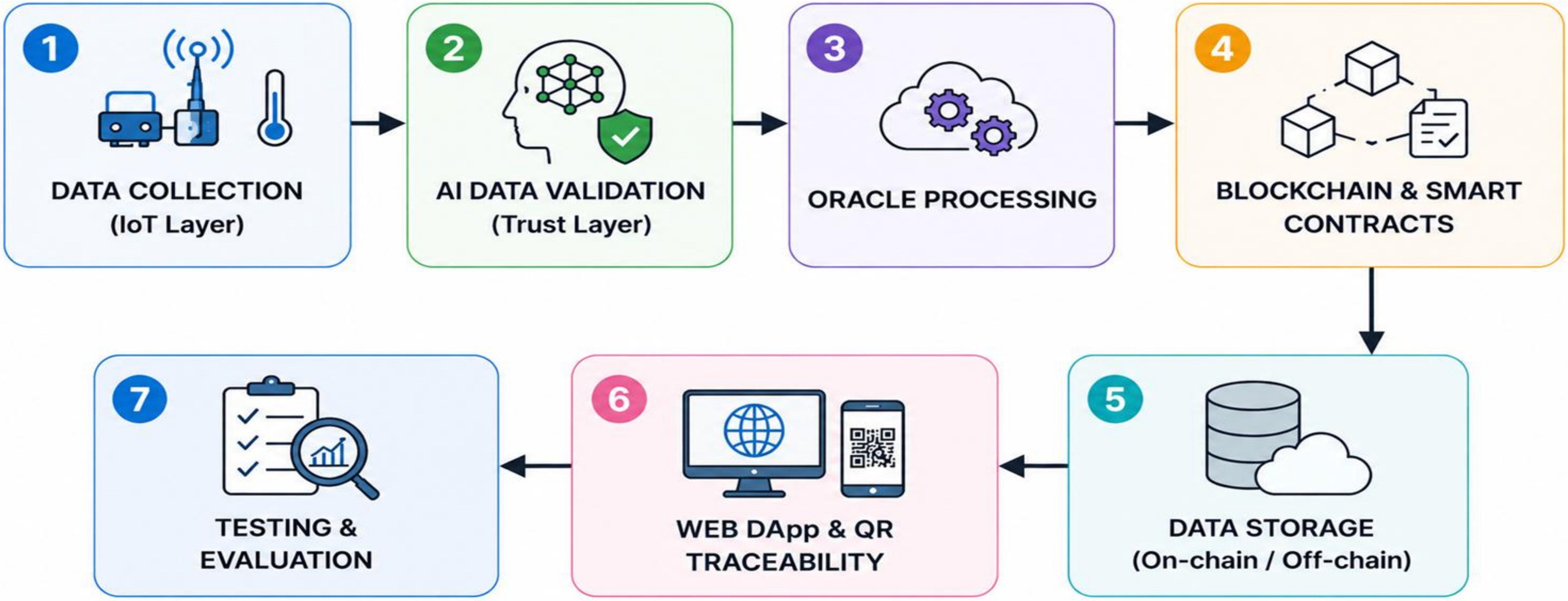
6. Farmer Payment: Enable transparent and rule-based payment processing so that farmers can receive their **proper and traceable payment**.

7. Monitoring & Verification: Provide stakeholders with access to trustworthy supply-chain information for verification and decision-making.

How It Is Better Than Existing Systems

Existing Systems	Proposed System
Mainly focus on blockchain-based traceability	Blockchain + IoT + AI + Smart Contracts
IoT data may be directly stored without validation	AI validates sensor data before recording
Limited customization of smart contracts	Customized smart contracts for supply-chain rules
Payment transparency may not be addressed	Transparent and traceable farmer payments
Separate technologies are often used independently	Integrated end-to-end framework
Greater risk of unreliable off-chain data	AI Trust Layer improves data reliability

Flowchart


1
DATA COLLECTION
(IoT Layer)
2
AI DATA VALIDATION
(Trust Layer)
3
ORACLE PROCESSING
4
BLOCKCHAIN & SMART
CONTRACTS
5
DATA STORAGE
(On-chain / Off-chain)
6
WEB DApp & QR
TRACEABILITY
7
TESTING &
EVALUATION

System Architecture

System Design — Simplified View

Four user roles · the physical goods flow · what the system records and verifies

USERS & PHYSICAL GOODS FLOW

🌱 Farmer
registers a harvest batch
crop, quantity, origin, date
sees consumer feedback ↩

→ goods →

🏭 Warehouse
accepts & stores the batch
logs storage conditions
confirms custody transfer

→ goods →

🏪 Dark store / Retailer
receives stock, manages
inventory & expiry
sells to consumer

→ goods →

👤 Consumer
scans QR → sees full journey
verifies origin & freshness
⭐ reviews the batch

⭐ review & rating loops back to the farmer

THE SYSTEM — every actor interacts through one web app

Web App (dApp)
role-based login
batch registration & handoff
inventory dashboard
QR scan → traceability view
review & rating form
Next.js

IoT Sensors
track the batch in transit
location + temperature
+ humidity, timestamps

simulated fleet
(+ optional 1 real device)

→

AI Trust Layer ★
checks every sensor reading
before it is recorded

genuine ✓ | suspicious ✗

bad data never gets stored permanently

→ data passes → only verified

Blockchain
permanent, tamper-proof record
- batch identity & origin
- who handed to whom, when
- condition & breach alerts
- smart contracts enforce rules

Solidity on Polygon Amoy

the app reads & writes the chain (role-gated)

WHAT IS STORED WHERE (hybrid design)

On the blockchain
batch identity & origin
custody transfers
condition/breach flags
hashes only — never bulk data
small, permanent, verifiable

Off-chain (Supabase / IPFS)
full sensor history
inventory & expiry tables
consumer review text
images, certificates (IPFS)
large, cheap, editable

THE CORE IDEA IN ONE LINE

A blockchain makes records **permanent — but permanence is worthless if the data was wrong.**

So we put an AI checkpoint in front of it, **so only trustworthy data gets recorded forever.**

Software & Hardware Requirements Specification

SOFTWARE & HARDWARE REQUIREMENT SPECIFICATIONS

SOFTWARE REQUIREMENTS

Operating System	:	Windows / Linux
Programming	:	Python, Solidity, JavaScript
Frontend	:	Next.js
Backend	:	FastAPI
AI/ML	:	Scikit-learn
Blockchain	:	Polygon Amoy
Smart Contracts	:	Solidity, Hardhat
Database	:	Supabase
Storage	:	IPFS
Wallet	:	MetaMask
Communication	:	MQTT / HTTPS

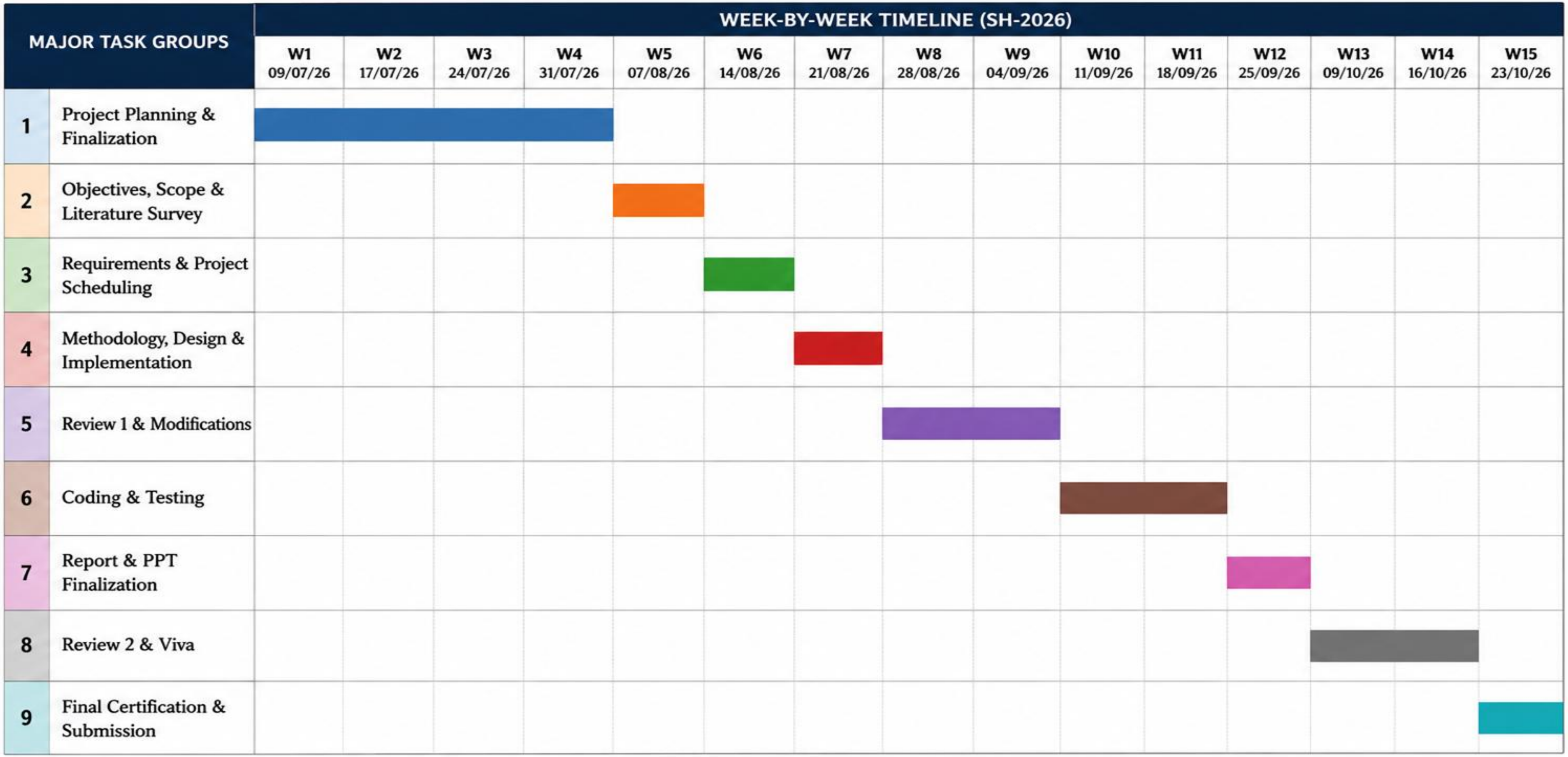
HARDWARE REQUIREMENTS

Processor	:	Intel i5 / Ryzen 5
RAM	:	8 GB minimum
Storage	:	256 GB SSD
Internet	:	Stable Internet connection
IoT Device	:	ESP32 / Raspberry Pi
Sensors	:	Temperature, Humidity, GPS
Mobile	:	Smartphone with camera for QR scanning

SYSTEM FLOW (OVERVIEW)

IoT Sensors → Python (Data Processing) → FastAPI (Backend) → Next.js (Frontend) → AI / ML (Scikit-learn) → Solidity (Smart Contracts) → Polygon Amoy (Testnet) → Supabase (Database) → IPFS (Storage)

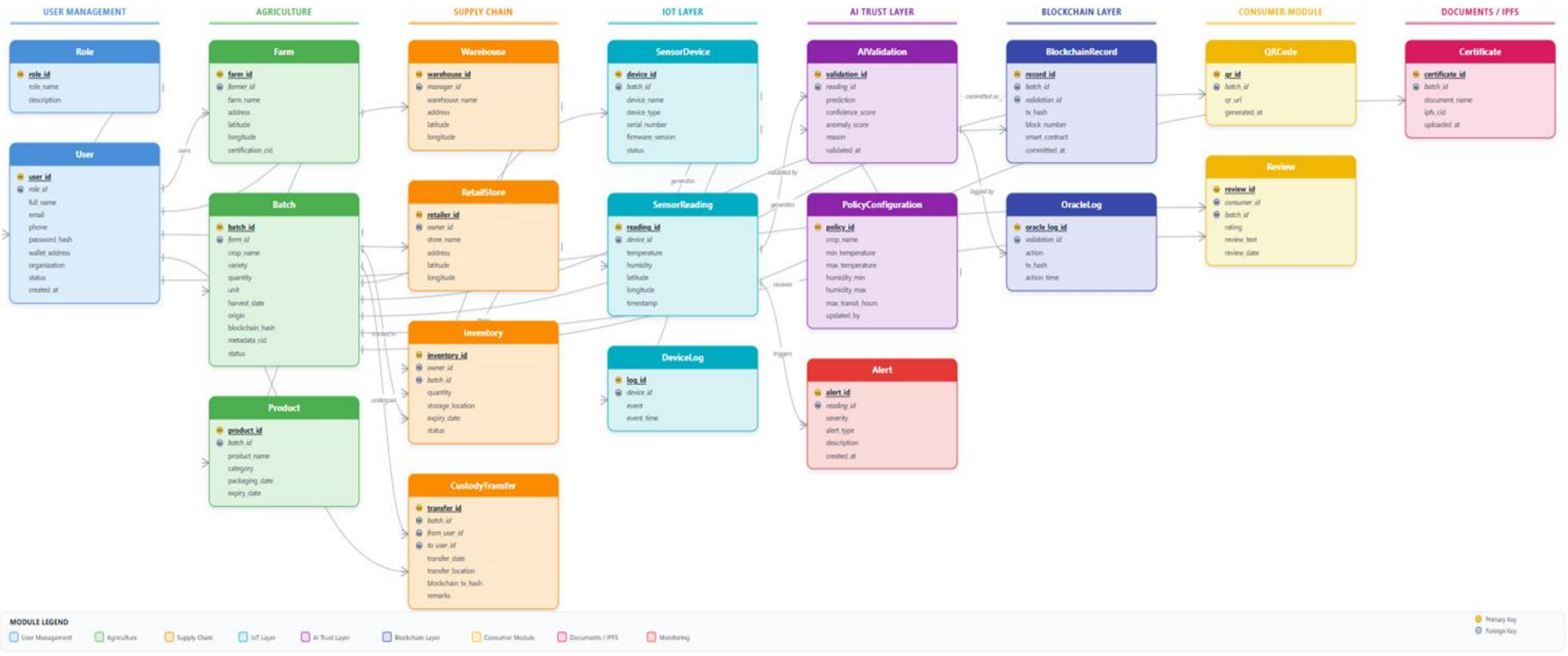
Project Scheduling / Timeline


MAJOR TASK GROUPS
WEEK-BY-WEEK TIMELINE (SH-2026)
W1 09/07/26
W2 17/07/26
W3 24/07/26
W4 31/07/26
W5 07/08/26
W6 14/08/26
W7 21/08/26
W8 28/08/26
W9 04/09/26
W10 11/09/26
W11 18/09/26
W12 25/09/26
W13 09/10/26
W14 16/10/26
W15 23/10/26
1 Project Planning & Finalization
2 Objectives, Scope & Literature Survey
3 Requirements & Project Scheduling
4 Methodology, Design & Implementation
5 Review 1 & Modifications
6 Coding & Testing
7 Report & PPT Finalization
8 Review 2 & Viva
9 Final Certification & Submission

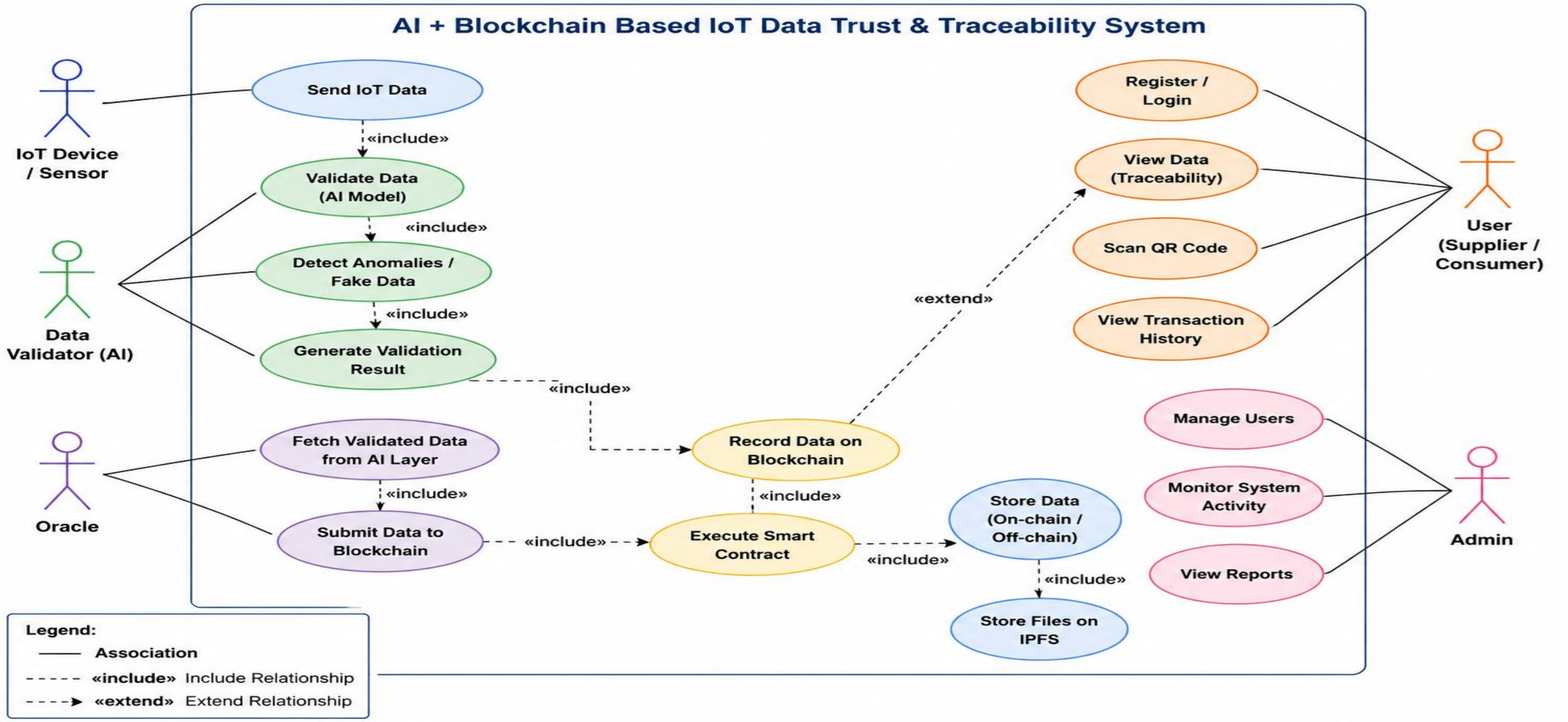
Entity Relationship Diagram (ERD)

Off-Chain Database Schema — Decentralized IoT-Blockchain Agricultural Traceability Platform

Entity Relationship Diagram (Crow's Foot Notation) • PostgreSQL / Supabase Layer



Use Case Diagram


AI + Blockchain Based IoT Data Trust & Traceability System
IoT Device / Sensor
Data Validator (AI)
Oracle
User (Supplier / Consumer)
Admin
Send IoT Data
«include»
Validate Data (AI Model)
«include»
Detect Anomalies / Fake Data
«include»
Generate Validation Result
«include»
Fetch Validated Data from AI Layer
«include»
Submit Data to Blockchain
«include»
Record Data on Blockchain
«include»
Execute Smart Contract
«include»
Store Data (On-chain / Off-chain)
«include»
Store Files on IPFS
«extend»
Register / Login
View Data (Traceability)
Scan QR Code
View Transaction History
Manage Users
Monitor System Activity
View Reports
Legend:
Association
«include» Include Relationship
«extend» Extend Relationship

Key Features

- **IoT-Based Real-Time Monitoring** – Collects agricultural supply-chain data such as product conditions, location, and movement.
- **AI Trust Layer** – Validates IoT data and detects abnormal or unreliable sensor readings before they are stored.
- **Blockchain-Based Traceability** – Maintains a tamper-resistant record of the product journey from **farmer to consumer**.
- **Customized Smart Contracts** – Automates quality verification, custody transfer, and payment-related rules.
- **Transparent Farmer Payments** – Provides a traceable and rule-based mechanism to ensure farmers receive their **proper payment**.
- **End-to-End Supply Chain Visibility** – Allows stakeholders to verify the history and movement of agricultural products.

Design & Implementation

AgriChain

Farm Fresh, Delivered With Trust

Handpicked produce from trusted farmers, delivered fresh to your doorstep.

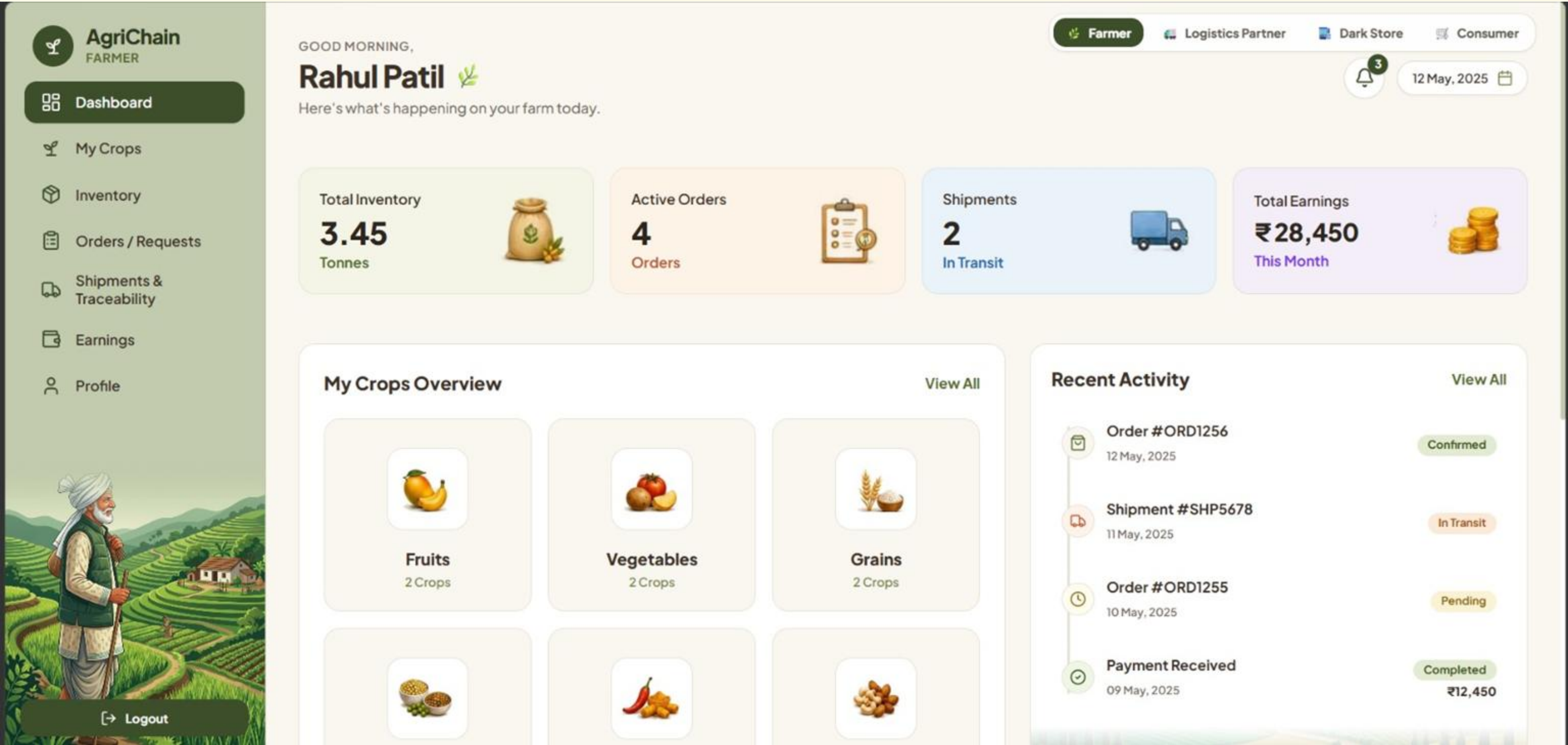
100% Natural
Pure, healthy and chemical free.

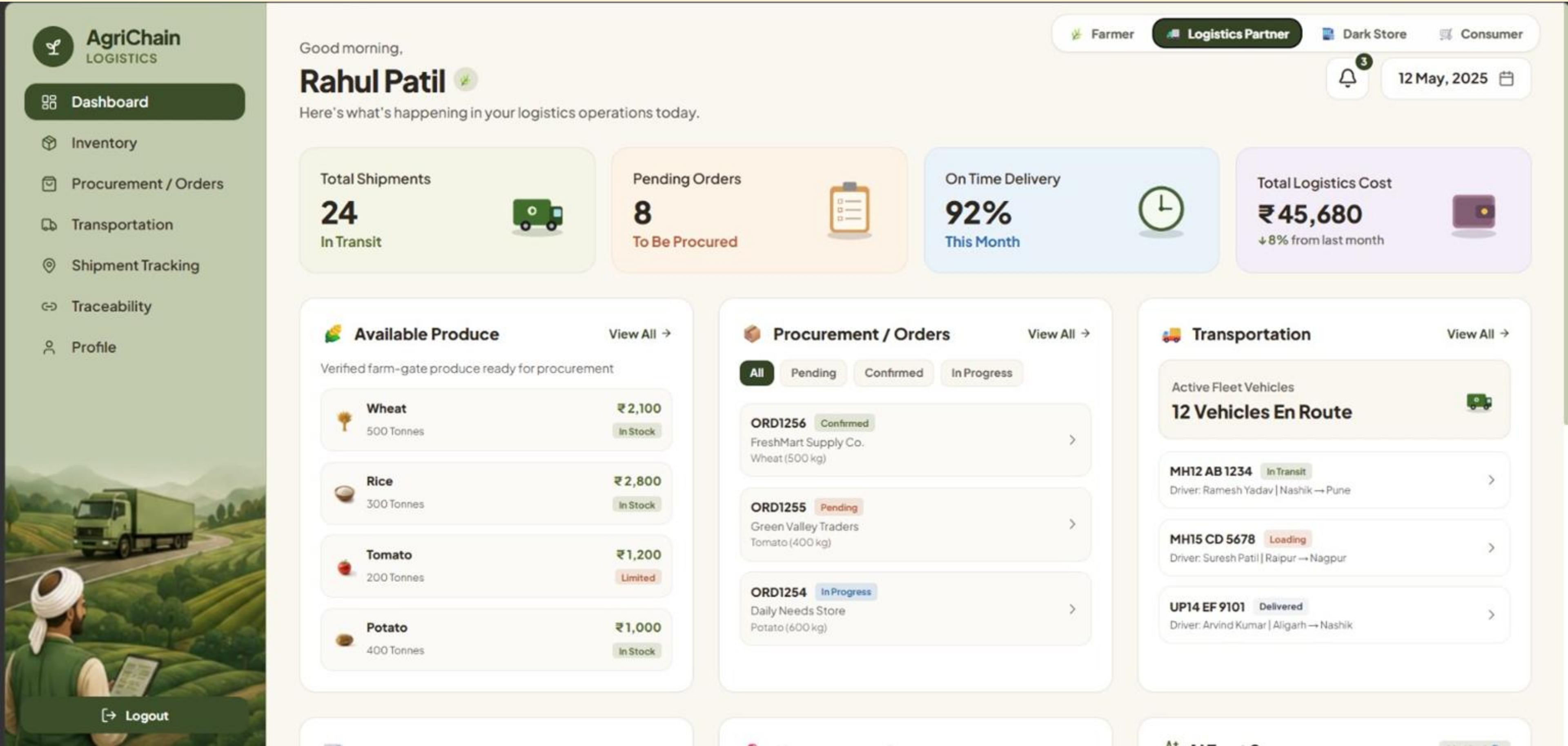
Trusted Farmers
Supporting local farmers and communities.

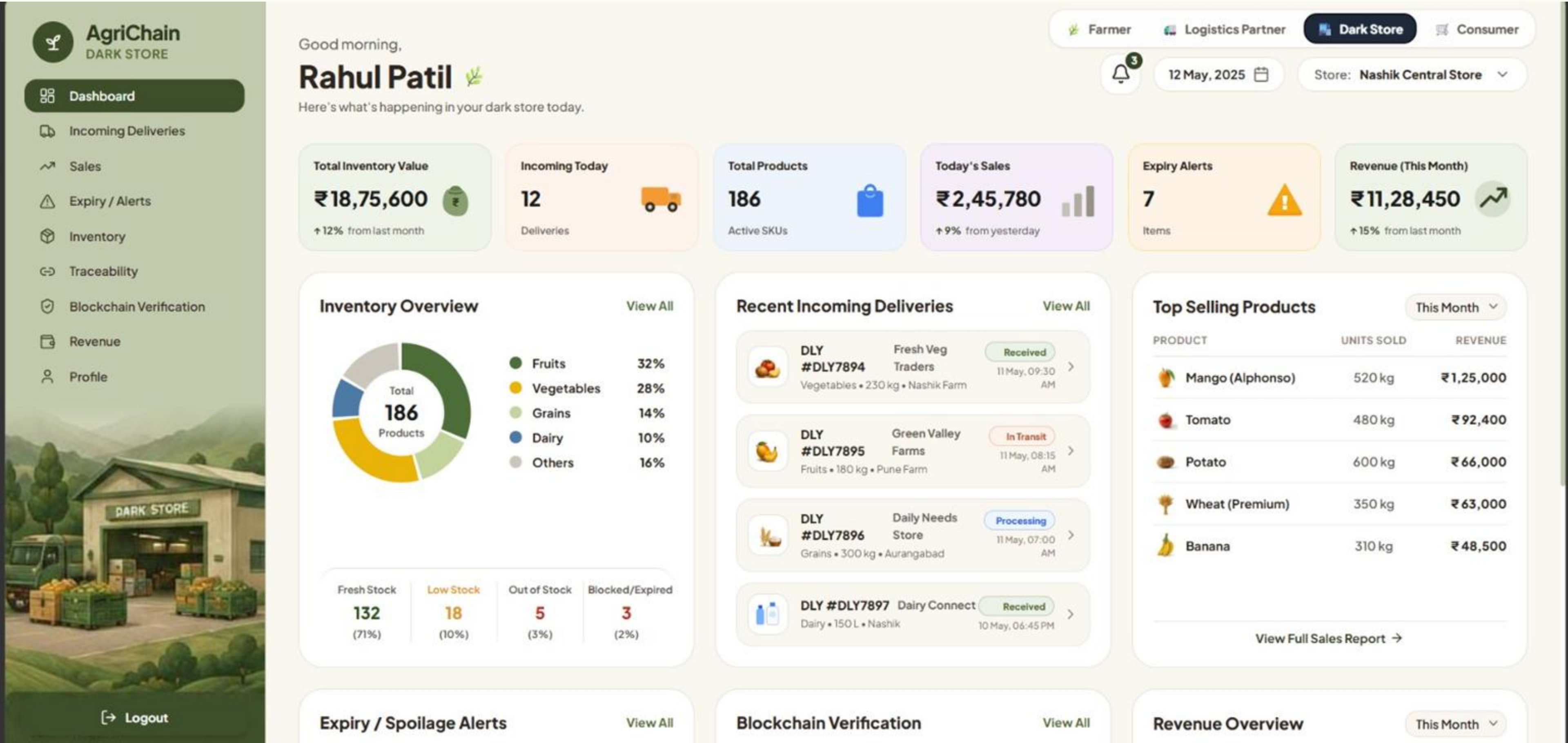
Safe & Secure
Hygienically packed and safely delivered.

Get Started →

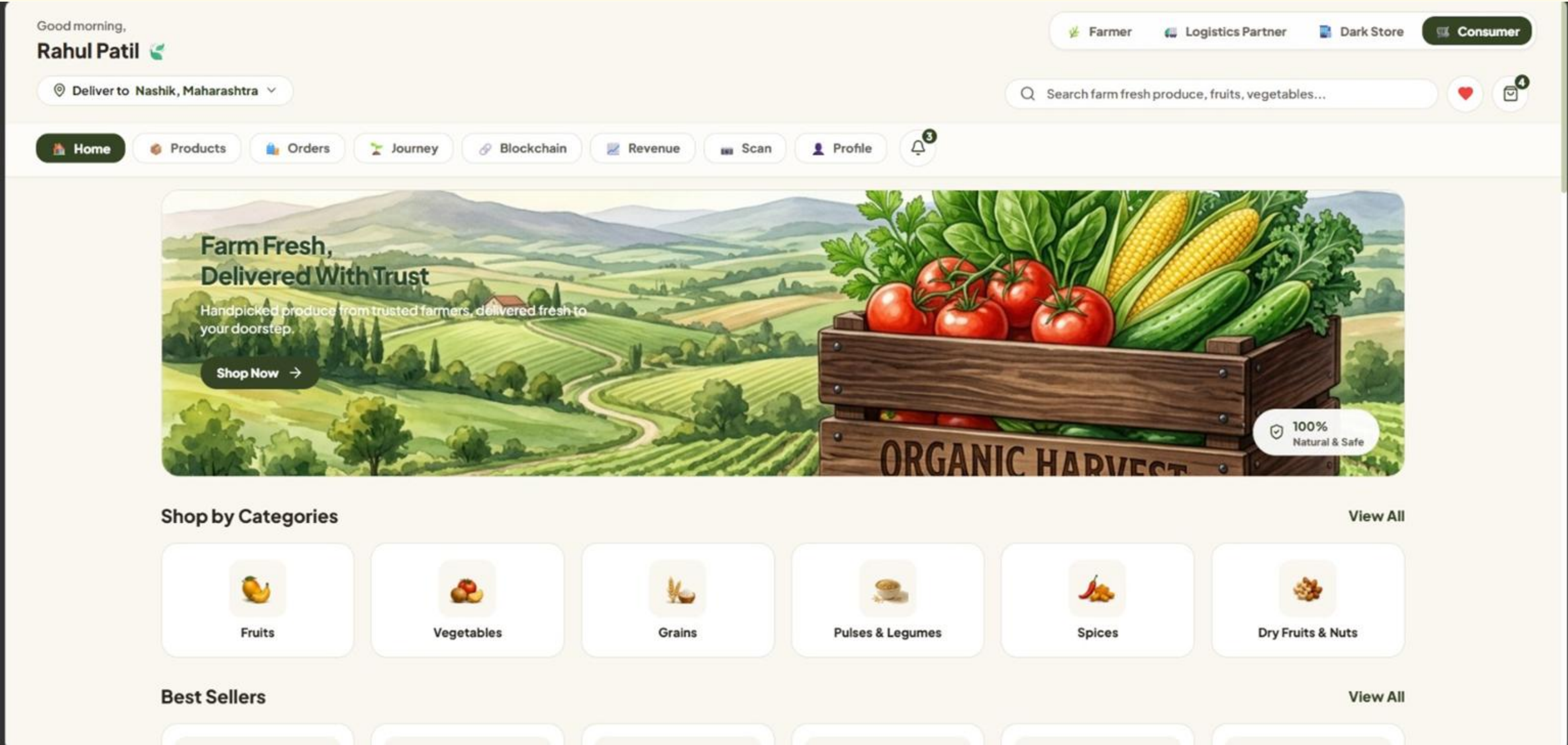
Continue as Guest


AgriChain
FARMER
Dashboard
My Crops
Inventory
Orders / Requests
Shipments & Traceability
Earnings
Profile
Logout
Farmer
Logistics Partner
Dark Store
Consumer
GOOD MORNING,
Rahul Patil
Here's what's happening on your farm today.
3
12 May, 2025
Total Inventory
3.45
Tonnes
Active Orders
4
Orders
Shipments
2
In Transit
Total Earnings
₹28,450
This Month
My Crops Overview
View All
Fruits
2 Crops
Vegetables
2 Crops
Grains
2 Crops
Recent Activity
View All
Order #ORD1256
12 May, 2025
Confirmed
Shipment #SHP5678
11 May, 2025
In Transit
Order #ORD1255
10 May, 2025
Pending
Payment Received
09 May, 2025
Completed
₹12,450


AgriChain
LOGISTICS
Dashboard
Inventory
Procurement / Orders
Transportation
Shipment Tracking
Traceability
Profile
Logout
Farmer
Logistics Partner
Dark Store
Consumer
Good morning,
Rahul Patil
Here's what's happening in your logistics operations today.
3
12 May, 2025
Total Shipments
24
In Transit
Pending Orders
8
To Be Procured
On Time Delivery
92%
This Month
Total Logistics Cost
₹45,680
↓8% from last month
Available Produce
View All →
Verified farm-gate produce ready for procurement
Wheat
500 Tonnes
₹2,100
In Stock
Rice
300 Tonnes
₹2,800
In Stock
Tomato
200 Tonnes
₹1,200
Limited
Potato
400 Tonnes
₹1,000
In Stock
Procurement / Orders
View All →
All
Pending
Confirmed
In Progress
ORD1256
Confirmed
FreshMart Supply Co.
Wheat (500 kg)
ORD1255
Pending
Green Valley Traders
Tomato (400 kg)
ORD1254
In Progress
Daily Needs Store
Potato (600 kg)
Transportation
View All →
Active Fleet Vehicles
12 Vehicles En Route
MH12 AB 1234
In Transit
Driver: Ramesh Yadav | Nashik → Pune
MH15 CD 5678
Loading
Driver: Suresh Patil | Raipur → Nagpur
UP14 EF 9101
Delivered
Driver: Arvind Kumar | Aligarh → Nashik


AgriChain
DARK STORE
Dashboard
Incoming Deliveries
Sales
Expiry / Alerts
Inventory
Traceability
Blockchain Verification
Revenue
Profile
DARK STORE
Logout
Farmer
Logistics Partner
Dark Store
Consumer
Good morning,
Rahul Patil
Here's what's happening in your dark store today.
3
12 May, 2025
Store: Nashik Central Store
Total Inventory Value
₹18,75,600
12% from last month
Incoming Today
12
Deliveries
Total Products
186
Active SKUs
Today's Sales
₹2,45,780
9% from yesterday
Expiry Alerts
7
Items
Revenue (This Month)
₹11,28,450
15% from last month
Inventory Overview
View All
Total
186
Products
Fruits 32%
Vegetables 28%
Grains 14%
Dairy 10%
Others 16%
Fresh Stock 132 (71%)
Low Stock 18 (10%)
Out of Stock 5 (3%)
Blocked/Expired 3 (2%)
Recent Incoming Deliveries
View All
DLY #DLY7894 Fresh Veg Traders Received
Vegetables • 230 kg • Nashik Farm 11 May, 09:30 AM
DLY #DLY7895 Green Valley Farms In Transit
Fruits • 180 kg • Pune Farm 11 May, 08:15 AM
DLY #DLY7896 Daily Needs Store Processing
Grains • 300 kg • Aurangabad 11 May, 07:00 AM
DLY #DLY7897 Dairy Connect Received
Dairy • 150 L • Nashik 10 May, 06:45 PM
Top Selling Products
This Month
PRODUCT UNITS SOLD REVENUE
Mango (Alphonso) 520 kg ₹1,25,000
Tomato 480 kg ₹92,400
Potato 600 kg ₹66,000
Wheat (Premium) 350 kg ₹63,000
Banana 310 kg ₹48,500
View Full Sales Report
Expiry / Spoilage Alerts
View All
Blockchain Verification
View All
Revenue Overview
This Month

Department Of Artificial Intelligence & Data Science



Conclusion

- The proposed system combines IoT, AI, blockchain, and customized smart contracts to create a transparent agricultural supply chain.
- The AI Trust Layer validates IoT data before it is recorded on the blockchain, improving the reliability of stored information.
- Blockchain and smart contracts provide tamper-resistant traceability, automated transactions, and transparent supply-chain records.
- The system aims to improve trust, accountability, and payment transparency for farmers while enabling stakeholders to verify the journey of agricultural produce.
- The proposed framework addresses key gaps identified in existing research, particularly off-chain data trust, traceability, and customized smart-contract-based supply-chain management.

References

[1] A. Agrawal et al., "Optimizing Supply Chain Management With IoE and AI," 2025.

[2] M. Mohamed, A. Al-Saleh, A. Sharma, and M. Tejani, "Zero-Day Exploits Detection with Adaptive WavePCA-Autoencoder (AWPA) Adaptive Hybrid Exploit Detection Network (AHEDNet)," Scientific Reports, 2025.

[3] A. Maurya et al., "Blockchain-Driven Security for IoT Networks: State-of-the-Art, Challenges and Future Directions," Peer-to-Peer Networking and Applications, 2025.

[4] D. Dudczyk, M. Dunston, and M. Crosby, "Blockchain Technology for Global Supply Chain Management: A Survey of Applications, Challenges, Opportunities and Implications," IEEE Access, 2024.

[5] S. Yavaprabhas, R. Kurnia, M. Seyedghorban, and D. Samson, "Demystifying the Impact of Blockchain on Trust in Emerging and Established Relationships: A Case of Organic Food Supply Chains," 2024.

Thank You